

Welcome to Code Crunch!

C12 | Crunch 3D: CAD & Modeling

[Tues 4-5PM] Weekly | Spring 2025

Attendance

Earn a Certificate of
Completion by attending
at least 70% of the
workshops



CODECRUNCH.CS.FIU



305 Hackathon : Edition Spring 2025



Funded by SGA
| Open to All University Students

305

SPRING 2025

HACKATHON

HYBRID

MAR 24 - 27 & MAR 29 - 31, 2025
WORKSHOPS | COMPETE | IN-PERSON CAREER FAIR

FIU
Honors College

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SIGN UP HERE: [CODECRUNCH.CS.FIU.EDU](https://codecrunch.cs.fiu.edu)

#CODECRUNCH-HACKS

CODE CRUNCH

OPEN TO ALL STUDENTS, REGARDLESS OF MAJOR
—NO EXPERIENCE NEEDED! FRESHMEN,
GRADUATES, AND ALUMNI FROM ANY UNIVERSITY
ARE WELCOME. LET'S INNOVATE TOGETHER!

FIU ORGS COLLABORATORS

Spring 2025 Weekly Schedule





Funded by SGA | Open to All Students

CODE CRUNCH WEEKLY SCHEDULE - SPRING 25

Your CS Club for Collaboration, Networking, and Hands-On Projects!
Open to all students, regardless of major—no experience needed! **Freshmen, graduates, and alumni from any university are welcome.** Let's innovate together!

C5 Crunch AI Data Science Python MON 1 - 2 PM WED 12:30 - 1:30 PM	C3 CodeCrunch LABS Portfolio Java & React MON 2 - 3 PM	C7 Wire Crunch: Embedded Systems C++ TUE 1 - 3 PM	C11 Crunch Arcade Unity TUE 3 - 4 PM FRI 4 - 5 PM	C12 Crunch 3D: CAD & Modeling OnShape TUE 4 - 5 PM	C9 Swe Convos Networking WED 1:30 - 2:30 PM FRI 2 - 3PM ***
C1 Crunch Convos Python WED 2:30 - 3:30 PM	C2 Crunch The Code Python WED 3:30 - 4:30 PM	C6 Cybersecurity Crunch VM/LINUX THU 1 - 2 PM	C4 CodeCrunch LABS Hackathon Java & React THU 2 - 3 PM	C8 CodeCrunch LABS: Web Dev Javascript THU 3 - 4 PM	C10 Crunching Bioinformatics Python FRI 3 - 4 PM

Attend 7 of 10 Weekly Sessions in the Same Workshop to Earn a Certificate of Completion!

RSVP on
LU.MA/CODECRUNCH

In person or
Stream on
Zoom



 CODECRUNCH.CS.FIU.EDU

 **January 6, 2025 - April 4, 2025**
Cost: No-cost

Become a member



Unit #4

Topic: Assemblies & Mating Constraints

C12 | Crunch 3D: CAD & Modeling

Key Dates and Structure

C12 - Crunch 3D: CAD & Modeling



- ★ **Week 1: Introduction to OnShape, Sketching Fundamentals, & 2D Constraints**

Learning what On-Shape is and how to properly sketch on OnShape.

- ★ **Week 2: 3D Part Modeling**

Learning how to use extrude, remove, fillet, chamfer, revolve, shell, and loft functions for 3D models.

- ★ **Week 3: 3D Part Modeling Continued**

Learning how to use draft, mirror, and sweep functions for 3D models.

- ★ **Week 4: Assemblies & Mating Constraints**

Learning how to create assemblies through mates and constraints.

- ★ **Week 5: Final Project - Real-World Design Challenge**

Applying all concepts to design a functional product.

- ★ **Week 6: Final Project Showcase**

Presenting your final project, sharing your experience, and receiving feedback.

Assembly Definitions

- **Assembly:** Allows to combine parts in different ways based on your needs.
- **Mate Connectors:** To make an assembly in OnShape, we use mate connectors. They are on the faces and edges of the parts, and we use them to connect the part we are working with.



Types of Mates

- **Fastened:** Combines the Mate Connectors and locks them in place. The icon for the Fastened mate is a grey cylinder with a white top and bottom.
- **Revolute:** Combines the Mate Connectors and only allows rotation about the Z-axis. The icon for the Revolute mate is a grey cylinder with two curved arrows on its side, indicating rotation.
- **Slider Mate:** Combines the Mate Connectors and only allows motion about the Z-axis. The icon for the Slider Mate is a grey cylinder with a vertical arrow on its side, indicating linear motion.
- **Planar Mate:** Combines the Mate Connectors and only allows motion about X and Y-axis and rotation about Z-axis. The icon for the Planar Mate is a square with a cross inside, indicating motion in the XY plane and rotation about the Z-axis.

Types of Mates

- **Cylindrical Mate:** Combines the Mate Connectors and allows motion and rotation about the Z-axis. The icon for a Cylindrical Mate, showing a circular arrow around a vertical axis.
- **Pin Slot Mate:** Combines the Mate Connectors and allows motion about the X-axis and rotation about the Z-axis. The icon for a Pin Slot Mate, showing a pin and a slot with arrows indicating motion.
- **Ball Mate:** Combines the Mate Connectors and allows rotation about the X, Y, and Z-axis. The icon for a Ball Mate, showing a sphere with arrows indicating rotation about multiple axes.
- **Parallel Mate:** Keeps the Mate Connectors parallel to each other. The icon for a Parallel Mate, showing two parallel lines with arrows indicating they remain parallel.
- **Tangent Mate:** Keeps the mate connectors tangent to each other. The icon for a Tangent Mate, showing two surfaces touching at a point with arrows indicating they remain tangent.



Session Overview

In this session we will be creating a **link assembly** to practice the different **mates**.

To make this assembly, we are going to need **two 3D models** of links. We will denote them as **Part 1** and **Part 2**. Then, we will put the models together through the **assembly**.

Time for...

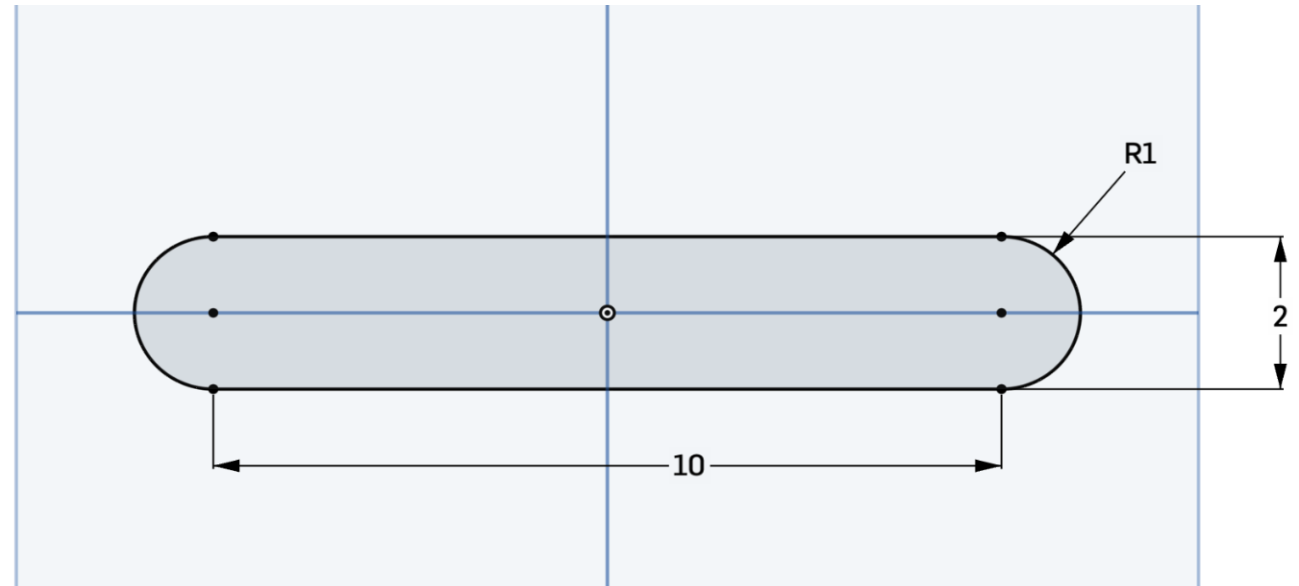


Part 1!

Part 1: Sketch 1

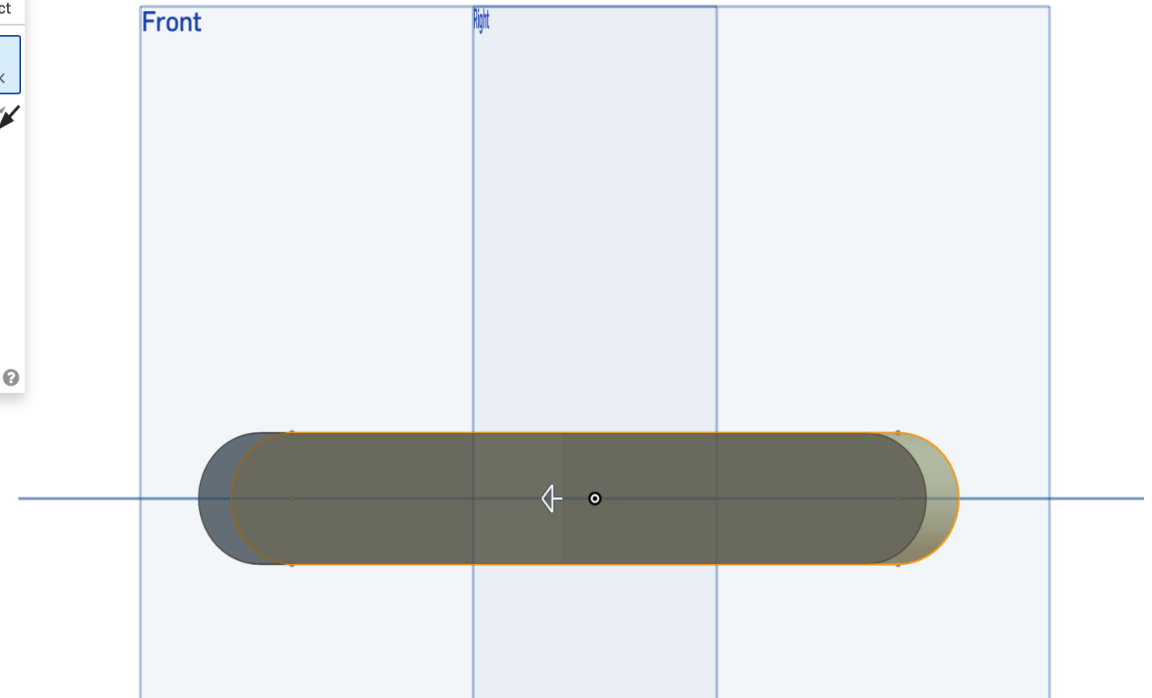
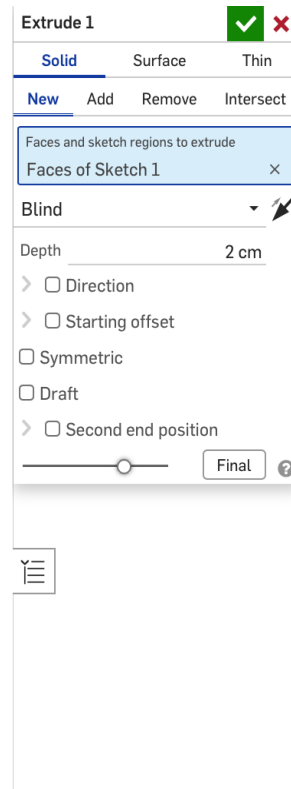
On the **Front Plane**, make a **Center Rectangle** with a **Tangent Arc** in each side.

The rectangle has a length 10 cm and the arcs with a radius of 1 cm.



Part 1: Extrude Base

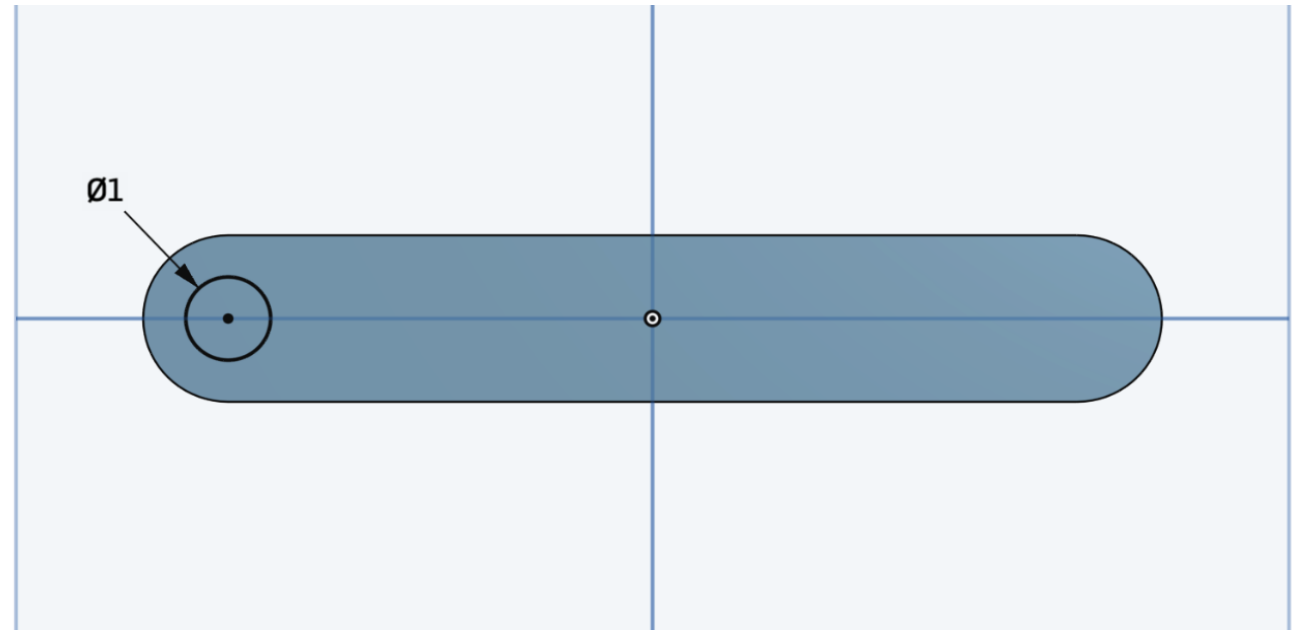
Make an **Extrude** of **Sketch 1** of 2 cm.



Part 1: Sketch 2

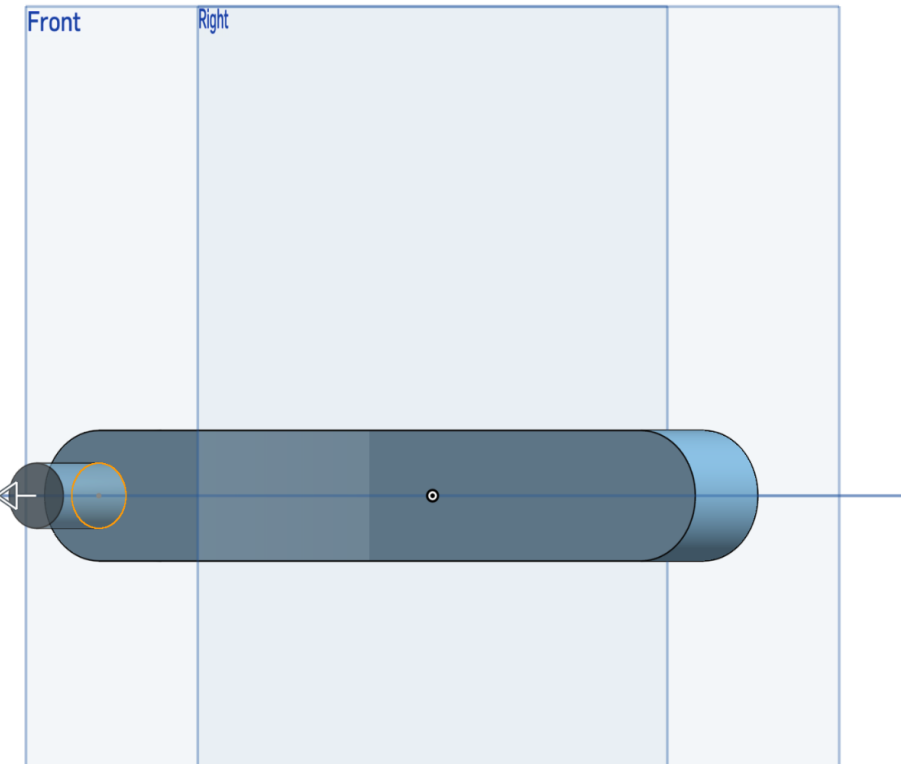
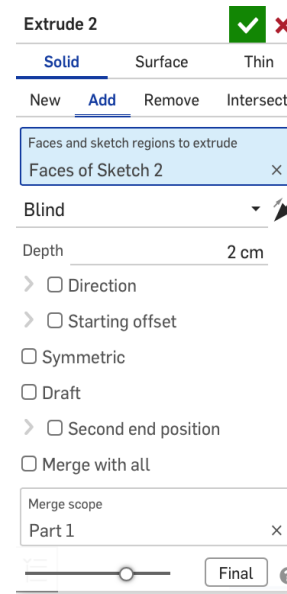
On the **Front Face** of the part, make a **Sketch**.

Draw a **Center Point Circle** on the center of the left arc of the model with a diameter of 1 cm.



Part 1: Extrude Boss

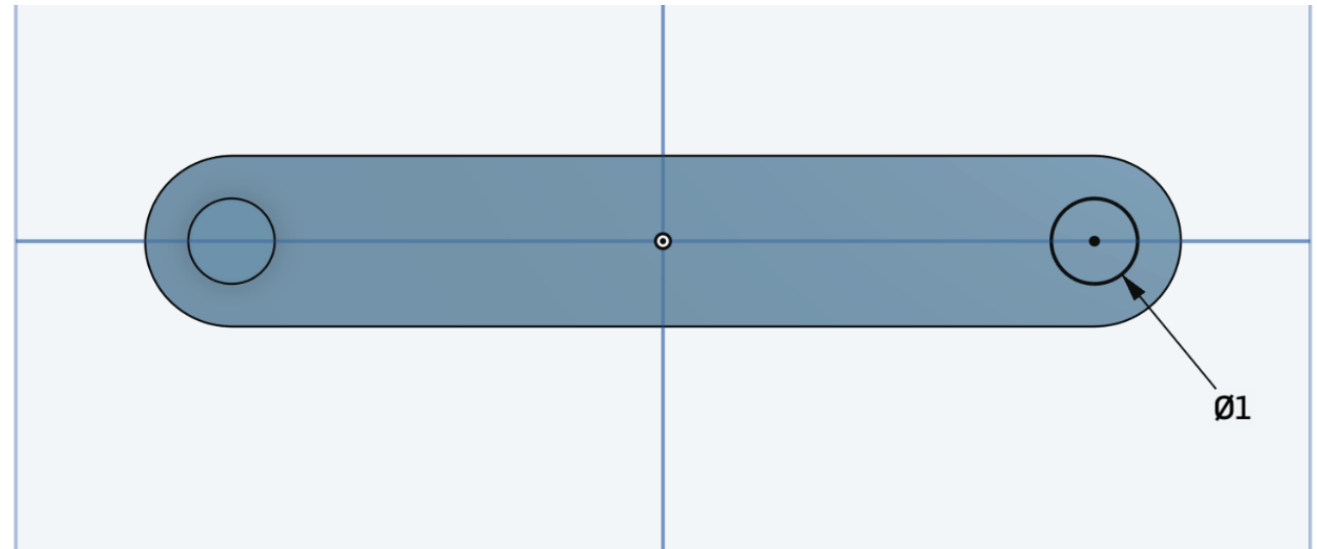
Make a **Extrude** of **Sketch 2** of 2 cm.



Part 1: Sketch 3

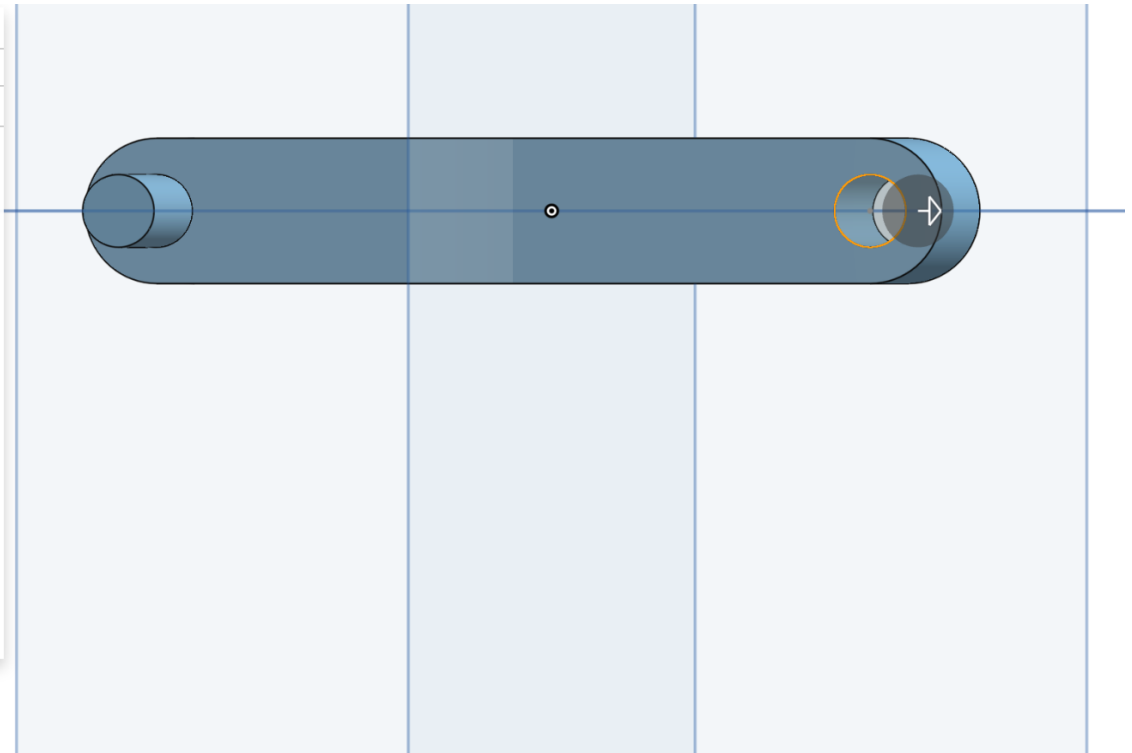
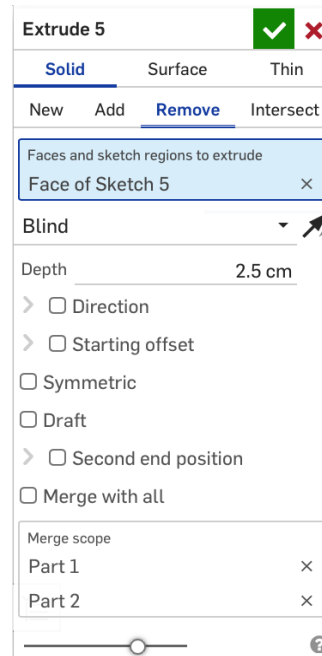
On the **Front Face** of the part, make a **Sketch**.

Draw a **Center Point Circle** on the center of the right arc of the model, with a diameter of 1 cm.



Part 1: Remove

Make a **Extrude**, removing area, of **Sketch 3**, making a hole on the part.



Time for...



Part 2!

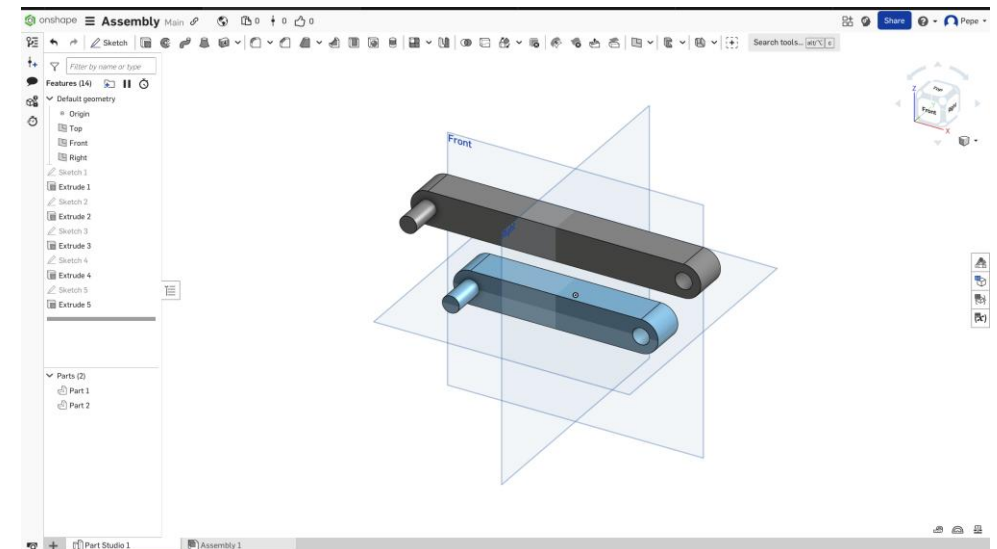
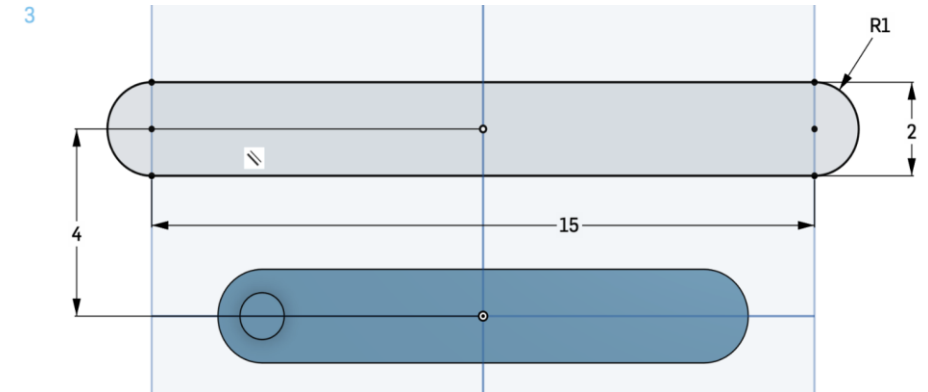
Part 2



OnShape allows the creation of multiple parts in one document or “**Part Studio**” tab by separating the parts.

To make **Part 2**:

- Create another **Sketch** in the **Front Plane**.
- Make the **center** of the Sketch above the **center** of the **Front Plane**. For the final sketch to be **fully defined**, make the distance 4 cm. This allows the new sketch to be separate from Part 1.
- Follow the same procedure as **Part 1**, but change the **length** of **Sketch 1** from 10 cm to 15 cm.

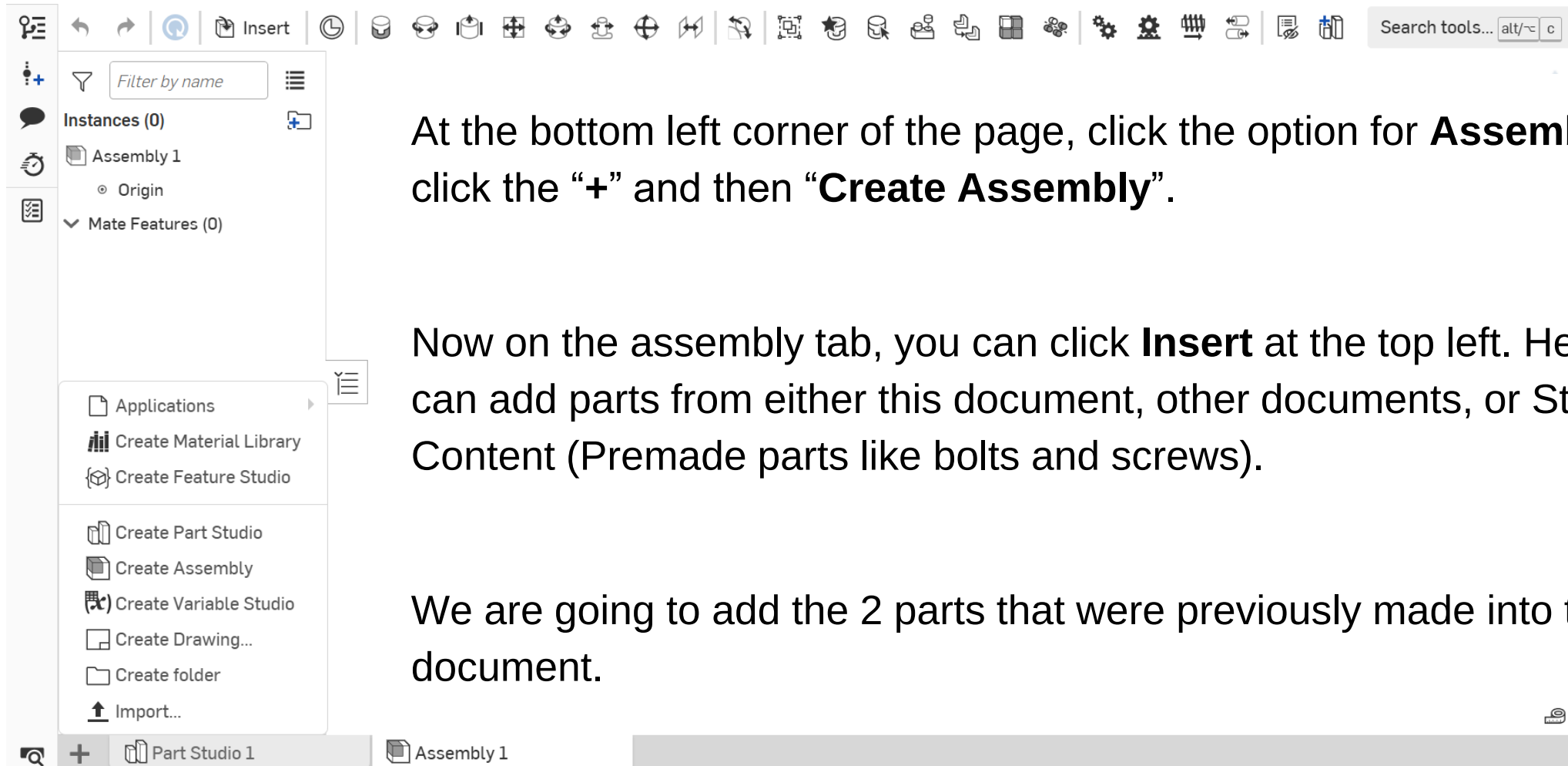




Time for the...

Link Assembly!

Assembly



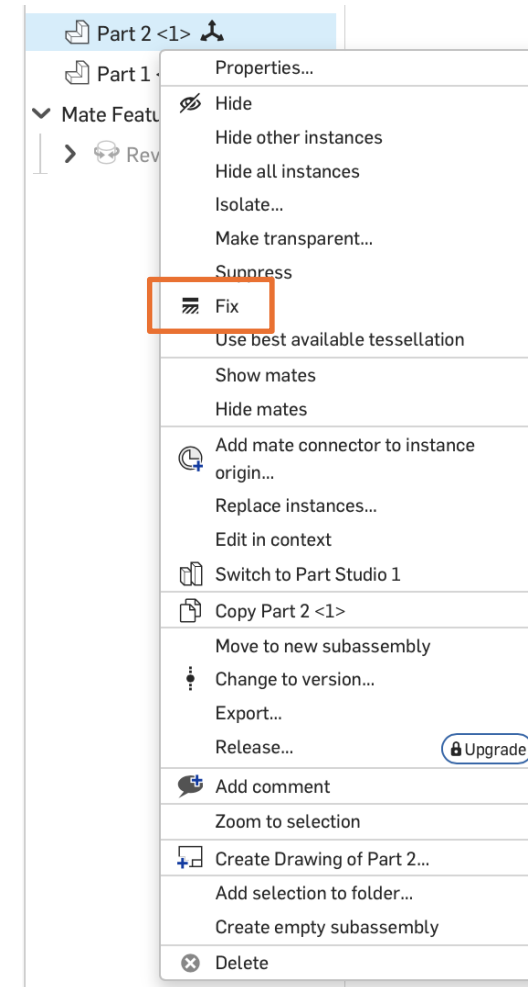
At the bottom left corner of the page, click the option for **Assembly 1** or click the “+” and then “**Create Assembly**”.

Now on the assembly tab, you can click **Insert** at the top left. Here you can add parts from either this document, other documents, or Standard Content (Premade parts like bolts and screws).

We are going to add the 2 parts that were previously made into this document.

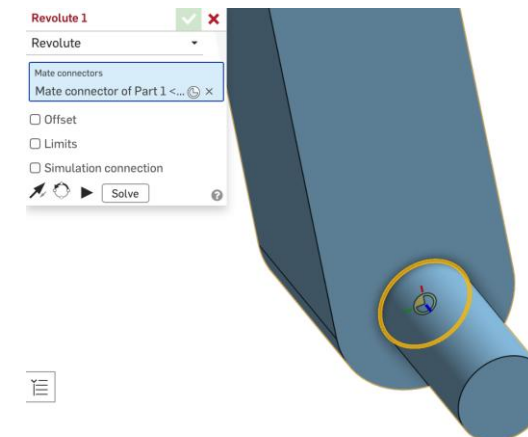
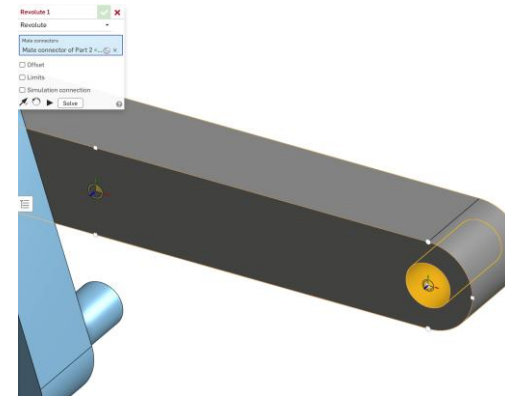
Links: Assembly

To make the **Assembly**, it is easier to have one piece as a “**base**”, meaning it **does not move**. To do this right click **Part 2** and select **Fix**.



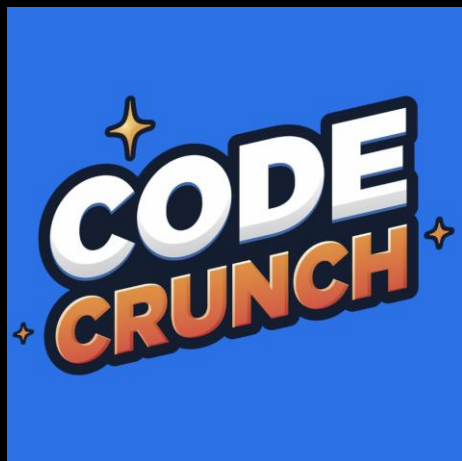
Links: Assembly

To **combine** these parts, we are going to use a **Revolute Mate**, allowing the links to **connect and rotate**. Combine the **Mate Connector** of the hole of **Part 2** with the **Mate Connector** at the bottom of the pin of **Part 1**.





Q&A Session



Attendance



Join us for the upcoming weekly sessions this Spring 2025!
(Jan 6 - Apr 4, 2025)



Monday: C5 1PM
Tuesdays: C7 1PM / C11 3PM
Wednesdays: C5 12:30 PM / C9 1:30 PM (Varies)
C1 2:30PM / C2 3:30PM / C13 5:30PM
Thursdays: C6 1PM / C8 3PM
Friday: C9 2PM (Varies) / C10 3PM/ C11 4PM



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